

Series 3: Applied Machine Learning and Predictive Modelling 1

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Load the `InsectSprays` dataset and have a look at the help page for it. Note that no add-on package needs to be loaded, this is a **R** “built-in” dataset.

```
data(InsectSprays)
help(InsectSprays)
```

Question

Use an appropriate function to visualise the data.

Question

What do you observe on the graph? Are there differences among groups? Is the variability within each group similar?

Question

Fit a Poisson model to the data and interpret the regression coefficients.

Question

Check whether the model is overdispersed. If this is the case, fit a quasipoisson model and compare the estimated regression coefficients with those of the Poisson model.

Question

*Formally test whether there is any evidence that the predictor **spray** plays a role in the quasipoisson model.*

Question

*We previously tested whether the predictor **spray** plays a role. Given that this test was statistically significant, we can dig further and test hypotheses that involve some of the levels of this predictor.*

Use the `glht()` function in `{multcomp}` package to test whether the conventional pesticides "C", "D" and "E" differ from the organic pesticides "A", "B" and "F".

Test also whether the newly developed pesticides “A” and “B” differ from the old conventional pesticide “F”. Use the quasipoisson model to test these hypotheses.

Question

Ask generative AI to

- *explain the differences between count and continuous data*
- *explain what model you should fit to count data*
- *explain what happens when you fit a normal linear model to count data*
- *interpret the coefficients you get from a Poisson model*

What can you share about these answers?